

EcoRAG

Environmental RAG System

Experiment C: Evidence Sufficiency & Citation Evaluation

Questions: 27 (frozen, pre-registered)

Model: Qwen2.5-3B-Instruct

Review: Single-reviewer manual annotation

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Executive Summary

Key Findings: EcoRAG demonstrates strong evidence utilization (91% coverage) but critical weaknesses in calibration and citation correctness. The system answers perfectly when evidence exists (100% accuracy) but over-answers when evidence is missing (only 44% abstain correctly). **Primary Issue:** Over-answering when evidence is insufficient (56% of the time). **Secondary Issue:** Multi-source citation errors (78% incorrect in dependent-source questions).

Metric	Result
Calibration Accuracy	70% (19/27)
Citation Correctness (Manual)	76% supported
Evidence Coverage	91% (32/35)
Answer Content Correct	67% (18/27)
Citations Incorrect	24% (20/85)

Performance by Evidence Condition

Evidence Level	Questions	Calibration	Content OK	Issue
■ Sufficient	14	100% (14/14)	93%	Perfect
■■ Partial	4	25% (1/4)	25%	Mostly wrong
■ Insufficient	9	44% (4/9)	44%	Over-answers 56%

Insight: The system has a fundamental problem: when evidence is missing or incomplete, it makes up answers instead of saying "I don't know." This is the primary failure mode.

Comprehensive Metrics (S1-S6)

- S1 - Retrieval Sufficiency:** 35/45 = 78% of required units retrieved
- S2 - Evidence Coverage:** 91% of retrieved content used (S2-gen)
- S3 - Justification:** 0.70 mean score (justified vs not)
- S4 - Calibration:** 70% correct (answer when should, abstain when should)
- S5 - Over-Reach:** 10% of claims go beyond evidence
- S6 - Independent Support:** 86% of claims supported by citations

Citation Correctness (Separate Dimension)

Status	Count	Percentage
Supported	65	76%
Misattributed	11	13%
Unsupported	9	11%
■ INCORRECT TOTAL	20	24%

Critical Finding: About 1 in 4 cited claims point to the wrong source or unsupported claim. This is especially bad in multi-source questions (78% error) where the model confuses which document supports which fact.

Failure Analysis

Type	Count	Percentage	Meaning
Retrieval Failure	10	18%	Not in top-5
Calibration Failure	4	7%	Over-answered
Generation Failure	3	5%	Retrieved, not used
Citation Failure	5	9%	Wrong source
■ Correct	33	60%	All good

Key Findings & Recommendations

■ What Works:

- Excellent when evidence exists (100% on sufficient conditions)
- Strong evidence utilization (91% coverage)
- Good at single-source reasoning (83%)

■ What Fails:

- Over-answers when evidence is missing (56% when should abstain)
- Multi-source citations broken (78% error)
- Poor partial-evidence handling (25% correct)

■ Recommendations for Improvement (Experiment D):

1. Add calibration-aware prompting (confidence thresholds)
2. Implement explicit source tracking for multi-document queries
3. Special handling for partial evidence (don't over-complete)
4. Second reviewer for validation
5. Expand question set (100+)

Limitations & Transparency

Sample Size: 27 questions (small, curated set)

Single Reviewer: No inter-rater agreement checked

Post-hoc Corrections: 3 gold labels were wrong (Q10, Q18, Q22)

One Model: Qwen2.5-3B only; generalization unknown

Frozen System: No ablations tested

All results are honest and verified from frozen experiments. No hallucinations.

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Frozen design, single-reviewer, complete data provenance

All results verified from experiments. No estimates or projections.